

Mustafa Jarrar

Full Professor

Computer Science, Birzeit University, Palestine

www.jarrar.info | mjarrar@birzeit.edu | +972-599-662258



- Director of **PhD Program in Computer Science**.
- Director of **SinaLab for Computational Linguistics and Artificial Intelligence**.
- Founder & Former Director of **Sina Institute for Knowledge Engineering and Arabic Technologies**.
- Founder & Director of the **Palestinian e-Government Academy**.
- CoFounder & Board member of the **Wiki Palestine Association**.
- Recent Achievements (Open Source): **Arabic lexicographic Database** | **Arabic Ontology** | **Palestinian Dialectal Corpus** | **Wojood NER** | **NLU Tools** | **Palestinian e-Government Framework**

Research Interests

Natural Language Processing, Social Computing, Ontology Engineering, Knowledge Graphs

Lexical Databases, Dialects, Social Media, AI Ethics, Hate Speech, Semantic Web, eGovernment, Interoperability, Linked-Data, Cultural Heritage, Upper Level Ontologies

Education

- **PhD in Computer Science** [2000 - 2005]
Vrije Universiteit Brussel, Belgium
Thesis: "Towards Methodological Principles for Ontology Engineering". Supervisor: Prof. Robert Meersman
- **Masters of Science in Computer Science** [1999 - 2000]
Vrije Universiteit Brussel, Belgium
- **Computer Systems Engineering** [1990 - 1995]
Palestinian Polytechnic University, Palestine

Employment

- **University of Birzeit**, Palestine. [2009 - present]
Professor of Computer Science
- **State University of New York at Buffalo**, USA. [2016 - 2017]
Research Associate Professor, Biomedical Informatics Department (Fulbright Scholar)
- **University of Cyprus**, Cyprus. [2007 - 2009]
Marie Curie Fellow, Computer Science Department
- **Vrije Universiteit Brussel**, Belgium. [1999 - 2007]
Senior Research Scientist
- **Arab Islamic Bank**, Palestine. [1998 - 1999]
Chief of Technology
- **Kinana for Computer & Engineering**, Palestine. [1996 - 1998]
Co-founder and CEO

Scientific Awards

1. **MIT Technology Review "Leading Arab Experts in AI"** (2022)
Top 30 Arab experts in AI worldwide
2. **Abdul Hameed Shoman Arab Researchers Award** (2019)
(\$20000) Technology - distinguished research on Arabic computational linguistics.
3. **Mohammed Bin Rashid Award for the Arabic Language** (2018)
(\$70000) Best intelligent applications - Lexicographic Search Engine.
4. **Best Poster Award** (2019)
The 2nd Conference on Language, Data and Knowledge.
5. **2nd Best Paper Award** (2017)
The 8th International Conference on Biomedical Ontology.
6. **Fulbright Visiting Scholar** (2016)
(1 year) spent at the State University of New York at Buffalo, USA.
7. **Google Faculty Research Award** (2016)
(\$51000) for developing an Arabic large-scale lexicographic database.
8. **The EU Marie Curie Fellowship** (2007)
(2 years) from the European Union, spent at the University of Cyprus.
9. **Best Paper Award** (2002)
The International Conference on Ontologies, Databases, and Applications of Semantics.
10. **BTC Doctoral Scholarship** (2000)
(3 years) from the Belgian Ministry of Cooperation, spent at the Free University of Brussels.

Research Grants

1. **CONVERSER** (Co-Principal Investigator)
(\$50000), TÜBİTAK-HCIE Program. [March 2021 - February 2023]
dialogue system (Chatbot) for Palestinian Banks that can understand standard and dialectal Arabic.
2. **Arabic Lemon** (Principal Investigator)
(\$16000) from **Birzeit University**. [December 2018 - December 2019]
To modernize and represent Arabic lexical resources using the W3C's Lemon Model.
3. **MOSIAC** (Partner)
(€798,000) from **European Union FP7, No. 612076**, 13 Partners, [2014 - 2015].
To foster R&D cooperation between the EU and Mediterranean countries on societal challenges.
4. **Arabic Ontology** (Principal Investigator)
(\$14500) seed funding from **Birzeit University**, [2010 - 2012].
To develop an Arabic Ontology; a formal specification of the concepts of all Arabic words. Similar to Arabic WordNet, but with ontological principles.
5. **VerbMesh** (Principal Investigator)
(\$11500) from **Birzeit University**. [July 2014 - January 2016]
To develop a large-scale graph of the Arabic verbs, and their inflectional relations.
6. **Curras** (Principal Investigator)
(\$35000) from the **Research Council- Ministry Of Higher Education**, [December 2013 - December 2015]
To develop a well-annotated corpus of the Palestinian Arabic dialect.
7. **Zinnar - The Palestinian e-Government Interoperability Framework** (Inventor)
Voluntary Consultancy for the Palestinian Government.
A framework to enable e-governmental and e-services in Palestine (adopted by all ministries).
8. **SIERA** (Coordinator)

- (€560,000) from **European Union FP7 (No. 295006)**, 5 Partners, [October 2011 - September 2014].
To build multilingual and multi-cultural knowledge sharing technologies.
9. **Palestinian e-Governance Academy (PalGov I)** (Coordinator)
(€564,964) from **the European Union TEMPUS (No. 511159)**, 11 partners, [2010 - 2012]
Capacity building for Palestinian ministries to gain the know-how for implementing e-government, in three main areas (interoperability, security, IT-laws).
 10. **GovSeer** (Partner)
(€50,000) from Italy CooperLink, 2 partners, [2010 - 2012].
Ontology-Based Interoperability of e-Government Services.
 11. **MashQL** (Principal Investigator)
(Non-funded project) to develop a novel approach for querying the Data Web (Web 3.0).
 12. **SEARCHiN** (Fellowship)
Funded by the European Union Marie Curie actions. 3 years. SEARCHing In a Networked World.
 13. **Knowledge Web** (Partner)
(€6,700,000) from the European Union FP6-IST-2002-507482 Networks of Excellence, [January 2004 - December 2007]
Supporting the transition of Ontology technology from Academia to Industry.
 14. **Innovanet** (Partner)
Funded the European Union FP5-IST-2001-38422.
Innovation engineering for the support of scientific discovery.
 15. **CCFORM** (Partner)
(€594,591) from the European Union FP5-IST-2001-34908 Networks of Excellence, [June 2002 - November 2003]
Consumer Complaint FORM for online resolution mechanisms.
 16. **OntoWeb** (Partner)
(€1,865,799) from the European Union FP5-IST-2000-29243 Thematic network, [June 2001 - May 2004]
Ontology-based Information Exchange for Knowledge Management and Electronic Commerce.
 17. **Namic** (Partner)
(€2,601,085) from the European Union FP5-IST-1999-12392 CSC, [March 2000 - April 2002]
News Agencies Multilingual Information Categorization.
 18. **IrisWeb**
Funded by DWTC Belgium
On ontology-based browsing in a (digital) library.
 19. **POLARIS**
Funded by Vrije Universiteit Brussel
To develop a prototype of the next-generation Digital University Library system.
 20. **DOGMA**
Funded by Vrije Universiteit Brussel
To develop Ontology-Guided Mediation of Applications. A framework for engineering and deploying ontologies.

Professional Memberships

1. **Palestinian National Curriculum Center Council.** Member (2024 - Present)
2. **Wiki Palestine Association.** Co-founder and Board Member (2022 - On)
3. **UN ESCWA Technology Centre (ETC) Board of Governors.** Board Member (2011 - 2014)
4. **National Committee for Science and Technology Statistics at the Palestinian Central Bureau of Statistics.** Member (2014 - Present)
5. **National E-Government Core Team.** Member (2010 - Present)
6. **National E-Government Interoperability Committee (Zinnar).** Founder and Member (2010 - Present)
7. **Marie Curie Alumni Association.** Member (2015 - Present)
8. **The Ontology Outreach Advisory.** Co-founder and Secretariat (2006 - 2012)
9. **IEEE Learning Technology Standards Committee.** Member (2005 - 2006)
10. **CEN/ISSS ICT Skills And Curricula.** Member (2006 - 2007)
11. **IFIP Working Group 2.6 On Databases.** Member (2003 - Present)

12. **IFIP Working Group 2.12 On Web Semantics.** Member (2004 - Present)
13. **The Palestine Engineers' Association.** Member (1996 - Present)

Roles at Birzeit University

1. Director: **PhD Program in Computer Science** (2020-On)
2. Director: **SinaLab for Computational Linguistics and Artificial Intelligence** (2011-On)
3. Chair: **University's Scientific Research Groups Committee** (2023-On)
4. Chair: **Master Students Scholarship Committee** (2022-On)
5. Founder and Director: **Sina Institute for Knowledge Engineering and Arabic Technologies** (2011 - 2015)
6. Director: **Master Program in Computing** (2010 - 2013)
7. Member: **University's Scientific Research Committee** (2010 - 2014, 2017 - 2018)
8. Member: **Faculty's Recruitment and Tenure Committee (CADER)** (2010 - 2015)
9. Member: Faculty Council of **Engineering and Technology** and **Graduate Studies and Research** (2010 - on)

Research Visits

1. **Institute of Technology, Villetaneuse, University of Paris 13**, France. Prof. Haifa Zargayouna. 3 weeks. Jun 2018
2. **State University of New York at Buffalo**, USA. Bioinformatics Dept., Prof. Werner Ceusters. 1 year. Jun 2017
3. **University of Namur**, Belgium. Faculty of Law, Prof. Romian Robert. 2 Weeks. Jun 2011
4. **University of Savoy**, France. Computer Science Dept. Prof. Christophe Roche. 1 Week. Jan 2011
5. **University of Trento**, France. Computer Science Dept. Prof. Marco Ronchetti. 2 Weeks. Nov 2010
6. **National & Kapodistrian University of Athens**, Greece. Computer Science Dept., Prof. Yannis Ioannidis. 1 week. Oct2008
7. **Free University of Bozen-Bolzano**, Italy. Computer Science Dept., Prof. Enrico Franconi. 3 Weeks. Oct 2006
8. **University of Trento**, Italy. Computer Science Dept. Prof. Fausto Giunchiglia. 3 Weeks. September 2006
9. **National Research Institute (CNR)**, Italy. Laboratory for Applied Ontology. Prof. Nicola Guarino. 2 Weeks. Jun 2006

Consultancy Services

1. **Board Member**, Arabica Encyclopedia - The Arab Center for Research and Policy Studies.
2. **Academic Evaluator**, promotion evaluations for 12 EU and Arab universities (Associate/Full Professor ranks).
3. **Arabic NLP Dialect Expert**, United Nations – NLP, Arabic dialects, and opinion mining in social media.
4. **Projects Evaluator**, for many universities and government programs.
5. **Proposal Evaluator**, European Union Erasmus+ Program - EACEA Agency.
6. **Proposal Evaluator**, European Union TEMPUS Program - EACEA Agency.
7. **Proposal Evaluator**, European Union Erasmus Mundus Program - EACEA Agency.
8. **Ministry Advisor**, E-Government Program - Palestinian Ministry of Telecommunication and IT.
9. **GIS and Spatial Databases Expert**, Palestinian Municipalities.
10. **Data Governance Expert**, Data-Driven Migration e-Governance, Palestinian Government.

Editor

1. Associate Editor, **Transactions on Asian and Low-Resource Language Information Processing**
2. Guest Editor, **Special Issue on Querying the Data Web -Novel techniques for querying structured data on the web**, Internet and Web Information Systems Journal (WWW), Springer. 2011
3. Guest Editor, **Special Issue on Knowledge Management and e-Human Resources Practices for Innovation**, International Journal of Knowledge and Learning, Inderscience. 2013

Organization of Conferences

1. Co-organizer, **FigNews: Framing the Israel War on Gaza: A Shared Task on News Media Narratives**, ArabicNLP 2024, Thailand.
2. Co-organizer, **AraFinNLP: Arabic Financial NLP Shared Task**, ArabicNLP 2024, Bangkok, Thailand.
3. Co-organizer, **ArabicNLU: Arabic Natural language Understanding Shared Task**, ArabicNLP 2024, Bangkok, Thailand.
4. Co-organizer, **WojoodNER: The 2st Arabic NER Shared Task**, ArabicNLP 2024, Bangkok, Thailand.
5. Co-organizer, **WojoodNER: The 1st Arabic NER Shared Task**, ArabicNLP 2023, Singapore.
6. Co-chair, SNAMS'18 Workshops **Fifth International Conference on Social Networks Analysis, Management & Security**, Spain.
7. Co-chair, Workshops, Tutorials, and Track 1 at the (AICCSA'18) **15th ACS/IEEE International Conference on Computer Systems and Applications**, Jordan.
8. Co-chair, Track 12 at the (AICCSA'17) **14th ACS/IEEE International Conference on Computer Systems & Applications**, Tunisia.
9. Co-chair, **Arab Big Data Initiative conference 2015**, Lebanon.
10. Chair, **Palestinian Big Data Initiative Workshop 2015**, Ramallah, Palestine.
11. Chair, (iArabic'15) **4th Palestinian Symposium on Computational Linguistics & Arabic Content**, Birzeit, Palestine.
12. Chair, (OnToContent 2015) **International Workshop on Ontology content & evaluation**, Rhodes Island, Greece.
13. Co-organizer, **Discussion Panel on Cross-Lingual Mapping Semantics**, at the 7th Global WordNet conference, Estonia.
14. Chair, (iArabic'14) **3rd Palestinian Symposium on Computational Linguistics & Arabic Content**, Birzeit, Palestine.
15. Chair, (OnToContent 2014) **International Workshop on Ontology content & evaluation**, Calabria, Italy.
16. Chair, (iArabic'13) **2nd Palestinian Symposium on Computational Linguistics & Arabic Content**, Birzeit, Palestine.
17. Chair, (PalGov'12) **1st National Conference on e-Governance & e-Services**, Birzeit, Palestine.
18. Chair, (OnToContent 2012) **International Workshop on Ontology content & evaluation**, Rome, Italy.
19. Co-organizer, (eHR-KM'11) **1st International Workshop on Knowledge Management & e-Human Resources Practices for Innovation**, Innsbruck, Austria.
20. Chair, (OnToContent 2010) **International Workshop on Ontology content & evaluation**, Crete, Greece.
21. Chair, (OnToContent 2009) **International Workshop on Ontology content & evaluation**, Algarve, Portugal.
22. Chair, (OnToContent 2008) **International Workshop on Ontology content & evaluation**, Monterrey, Mexico.
23. Chair, (OnToContent 2007) **International Workshop on Ontology content & evaluation**, Albufeira, Portugal.
24. Chair, (OnToContent 2006) **International Workshop on Ontology content & evaluation**, Montpellier, France.
25. Chair, (OOA-HR 2006) **Semantic challenges & opportunities in the Human Resources Domain**, Oxford, UK.
26. Chair, (OnToHR 2007) **International Workshop on Semantics Meets Human Resources and Employability**, Maastricht, Netherlands.
27. Co-chair, (BADE 2007) **IEEE Workshop on Biomedical Applications for Digital Ecosystems**.
28. Co-chair, (SWWS 2006) **IFIP WG 2.12 and WG 12.4 Workshop on Web Semantics**, Montpellier, France.
29. Co-chair, (WORM 2005) **3rd International Workshop on Regulatory Ontologies**, Larnaca, Cyprus.
30. Co-chair, (SWWS 2005) **IFIP WG 2.12 and WG 12.4 Workshop on Web Semantics**, Larnaca, Cyprus.
31. Co-chair, (WORM 2004) **2nd International Workshop on Regulatory Ontologies**, Larnaca, Cyprus.
32. Chair, (OMAC 2004) **Workshop on Ontology Modularization and context**, Brussels, Belgium.
33. Co-chair, (WORM 2003) **1st International Workshop on Regulatory Ontologies**, Catania, Sicily, Italy.

Awards Committees

1. **ESCWA Digital Arabic Content Award**, 2023
2. **ESCWA Digital Arabic Content Award**, 2022
3. **Arabic Hackathon**, 2021
4. **NYUAD Hackathon for Social Good in the Arab World**, 2019

Referee in Journals

1. Springer book publishing proposals.
2. **International Journal of Speech Technology**, Springer.
3. **Journal of Information Systems**, Elsevier.
4. **Journal of Applied Ontology**, IOS.
5. **Journal of Knowledge Engineering Review**, Cambridge.
6. **Journal of Data and Knowledge Engineering Journal**, Elsevier.
7. **Journal of Logical and Algebraic Methods in Programming**, Elsevier.
8. **Journal IEEE Transactions on Knowledge and Data Engineering**, IEEE.

9. Journal Advances in Engineering Software, Special Issue on Vocabularies, Ontologies and Rules, Elsevier.
10. IEEE Transactions on Industrial Electronics, IEEE.
11. Journal of Visual Languages and Computing, Special Issue on Perception and ontologies in visual, virtual and geographic spaces, Elsevier.
12. Journal on Web Semantics, Elsevier.
13. Journal of IEEE Software, IEEE.
14. Journal of Semantic Web and Information Systems, IGI-Global.
15. Journal of Electronic Commerce Research and Applications, special issue on Semantic Web Rules, Elsevier.
16. Journal on Data Semantics, Springer.
17. Semantic Web journal, IOS.
18. Journal of King Saud University, Elsevier.
19. Frontiers in Artificial Intelligence, frontiersin.
20. Journal of Artificial Intelligence Tools, World Scientific.
21. International Arab Journal of Information Technology (IAJIT), Zarka Private University.
22. Handbook of Research on Emerging Rule-Based Languages and Technologies, IGI-Global.
23. Special Issue of Fundamenta Informaticae for CILC 2011, the annual meeting of the Italian Association for Logic Programming, Association for Logic Programming.
24. Business Intelligence Book, Springer.

Program Committee Member in Conferences

1. ArabicNLP'24, The 2nd Arabic Natural Language Processing Conference, Thailand.
2. LREC-COLING 2024, The 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation, Torino Italy.
3. ESWC'24, The Extended Semantic Web Conference, Heraklion, Greece.
4. ArabicNLP'23, The 1st Arabic Natural Language Processing Conference, Singapore.
5. ISWC'23, The 22th International Semantic Web Conference . Greece
6. ESWC'23, The Extended Semantic Web Conference, Heraklion, Greece.
7. GWC'23, The 12th International Global WordNet Conferences, San Sebastian (Spain).
8. WANLP'22, The 7th Arabic Natural Language Processing Workshop, Abu Dhabi, UAE
9. LREC'22, The 13th International Conference on Language Resources and Evaluation, Palais du Pharo, France.
10. ISWC'22, The 20th International Semantic Web Conference . Hangzhou, China
11. ESWC'22, The Extended Semantic Web Conference, Heraklion, Greece.
12. OSCAT'22, The 5th Workshop on Open-Source Arabic Corpora and Processing Tools, Marseille, France.
13. Senior PC Member: ISWC'21, The 20th International Semantic Web Conference .
14. ESWC'21, The Extended Semantic Web Conference, Heraklion, Greece.
15. ACLing'21, 5th International Conference on Arabic Computational Linguistics.
16. WANLP'21, The 6th Arabic Natural Language Processing Workshop, EACL 2021 (Virtual)
17. ESAS'21, The 16th IEEE International Workshop on E-Health Systems and Web Technologies, Virtual
18. SIMPDA'21, The 11th IFIP International Symposium on Data-Driven Process Discovery and Analysis, Estonia.
19. Senior PC Member: ISWC'20, The 19th International Semantic Web Conference , Athens, Greece.
20. LREC'20, The 12th International Conference on Language Resources and Evaluation, Marseille, France.
21. ESWC'20, The Extended Semantic Web Conference, Crete, Greece.
22. WANLP'20, The 5th Arabic Natural Language Processing Workshop, Barcelona, Spain.
23. ACL'19, The 57th Annual Meeting of the Association for Computational Linguistics, Florence, Italy.
24. Senior PC Member: ISWC'19, The 18th International Semantic Web Conference , Auckland, New Zealand.
25. ESWC'19, The Extended Semantic Web Conference, Portoroz, Slovenia.
26. WI'19, The 18th Web Intelligence conference (WI'19), Thessaloniki, Greece.
27. ICALP'19, The 6th International Conference on Arabic Language Processing, Nancy, France.
28. WANLP'19, The 4th Arabic Natural Language Processing Workshop, Florence, Italy.
29. DiLiTAL'2019 , Diversite and TAL Linguistique, Oujda, Morocco.
30. NLP-DH2019, The first international workshop on NLP for Digital Humanities, Algeria.
31. SIMPDA'19, The IFIP International Symposium on Data-Driven Process Discovery and Analysis, Bled, Slovenia.
32. SemBDM'19, The IFIP Semantics in Big Data Management.
33. SNAMS'19, The Sixth International Conference on Social Networks Analysis, Management & Security, Granada, Spain.
34. ISWC'18, The 17th International Semantic Web Conference , California, USA.
35. AICCSA'18, The 15th ACS/IEEE International Conference on Computer Systems and Applications, Aqaba, Jordan.
36. SNAMS'18, The Fifth International Conference on Social Networks Analysis, Management & Security, Valencia, Spain.

37. ACLing'18, 4th International Conference on Arabic Computational Linguistics, Dubai, UAE.
38. MISC18, International Conference on Modern Intelligent Systems Concepts, Rabat, Morocco.
39. AICCSA'17, The 14th ACS/IEEE International Conference on Computer Systems and Applications, Hammamet, Tunisia.
40. WWW'17, The 2017 World Wide Web conference, Perth, Australia.
41. ISWC'17, The 16th International Semantic Web Conference , Vienna, Austria.
42. ESWC'17, The 14th Extended Semantic Web Conference, Portoroz, Slovenia.
43. ACLing'17, 3rd International Conference on Arabic Computational Linguistics, Dubai, UAE.
44. ICALP'17, The 6th International Conference on Arabic Language Processing, Fez, Morocco.
45. ICTCS'17, The International Conference on New Trends in Computing Sciences, Amman, Jordan.
46. AISI'17, International Conference on Advanced Intelligent Systems and Informatics, Cairo, Egypt.
47. WANLP'17, The 3rd Arabic Natural Language Processing Workshop, Valencia, Spain.
48. FBM'17, International Workshop on Fact Based Modeling, Rhodes, Greece.
49. SIMPDA'17, The IFIP International Symposium on Data-Driven Process Discovery & Analysis, Switzerland.
50. PICICT'17, The 2nd Palestinian International Conference on Information & Communication Technology, Gaza, Palestine.
51. ESAS'17, The 12th IEEE International Workshop on E-Health Systems and Semantic Web, Torino, Italy.
52. ESWC'16, The 13th European Semantic Web Conference, Crete, Greece.
53. AISI'16, International Conference on Advanced Intelligent Systems and Informatics, Cairo, Egypt.
54. SOTICS'16, The 6th International Conference on Social Media Technologies, Communication, and Informatics, Brussels, Belgium.
55. PAIS'16, Conference on Prestigious Applications of Intelligent Systems, The Hague, The Netherlands.
56. SIMPDA'16, The IFIP International Symposium on Data-Driven Process Discovery and Analysis, Graz, Austria
57. FBM'16, International Workshop on Fact Based Modeling, Rhodes, Greece.
58. IWOOD'16, 4th International Workshop on Definitions in Ontologies, Corvallis, Oregon, USA.
59. ESAS'16, The 11th IEEE International Workshop on E-Health Systems and Semantic Web, Georgia, USA.
60. ISWC'15, The 14th International Semantic Web Conference , Bethlehem, Pennsylvania, USA.
61. ESWC'15, The 12th European Semantic Web Conference, Portoroz, Slovenia.
62. SIMPDA'15, The IFIP International Symposium on Data-Driven Process Discovery and Analysis, Vienna, Austria.
63. PICCIT'15, The 4th Palestinian International Conference on Computer & Information Technology, Hebron, Palestine.
64. ICIT'15, The 7th International Conference on Information Technology, Amman, Jordan.
65. WCIS'15, The International Workshop on Computers and Information Sciences, Tabuk, KSA.
66. ESAS'15, The 10th IEEE International Workshop on E-Health Systems and Semantic Web, Taichung, Taiwan
67. ISWC'14, The 13th International Semantic Web Conference, Italy.
68. TOTH'14, The 8th TOTH International Conference on Terminology and Ontology: Theories and applications
69. CITALA'14, The 5th International Conference in Arabic Language Processing
70. ANLP'14, workshop on Arabic Natural Language Processing at EMNLP 2014
71. SIMPDA'14, The IFIP International Symposium on Data-Driven Process Discovery and Analysis
72. ORM'14, Workshop on Fact-Oriented Modeling.
73. ESAS 2014: The 9th IEEE International Workshop on Engineering Semantic Agent Systems
74. SWWS 2014: 10th International IFIP Workshop on Semantic Web & Web Semantics
75. ISWC'13, The 12th International Semantic Web Conference
76. ORM'13, Workshop on Fact-Oriented Modeling.
77. SIMPDA'13, The IFIP International Symposium on Data-Driven Process Discovery and Analysis
78. SIMPDA'13, The IFIP International Symposium on Data-Driven Process Discovery and Analysis (Asian Edition)
79. TOTH'13, The 7th TOTH International Conference ?? Terminology & Ontology: Theories and applications
80. COLING'12, The 2013 International Conference on Terminology & Ontology : Theories and applications
81. CITALA'12, The 4th International Conference on Arabic Language Processing
82. SIMPDA'12, The IFIP International Symposium on Data-Driven Process Discovery and Analysis
83. ORM'12, Workshop on Fact-Oriented Modeling.
84. ColabTKR, 2012 - Collaboration in Terminology and Knowledge Representation.
85. ESAS 2012: The 7th IEEE International Workshop on Engineering Semantic Agent Systems
86. SWWS'12, 8th International IFIP Workshop on Semantic Web & Web Semantics
87. META4eSociety'12, Methods, Evaluation, Tools and Applications for the Creation and Consumption of Structured Data for the e-Society
88. SIMPDA'11, The IFIP International Symposium on Data-Driven Process Discovery and Analysis
89. RuleML'11-USA, The International Symposium on Rules, RuleML.
90. RuleML'11_EU, The International Symposium on Rules, RuleML.
91. ISWSA'11, The International Conference Intelligent Semantic Web.
92. ORM'11 Workshop on Fact-Oriented Modeling.
93. INBAST'11, Workshop on Industrial and Business Applications of Semantic Web Technologies
94. SLSIS'11 International Workshop on Survivable Large-Scale Information Systems.
95. ESAS'11, 4th IEEE International Workshop on Engineering Semantic Agent Systems.
96. KARE'10, The 2nd International Workshop on Knowledge Acquisition, Reuse and Evaluation.

97. ISWSA'10, The International Conference Intelligent Semantic Web.
98. VORTE'10, International Workshop on Vocabularies, Ontologies and Rules for The Enterprise.
99. LOAIT'10, International Workshop on Legal Ontologies and Artificial Intelligence Techniques.
100. PICCIT'10, The 3rd Palestinian International Conference on Computer and IT.
101. ORM'10 Workshop on Fact-Oriented Modeling.
102. SWWS'10 6th International IFIP Workshop on Semantic Web & Web Semantics
103. ISWC'09, The International Semantic Web Conference -Demo and Posters track.
104. KEOD'09, The International Conference on Knowledge Engineering and Ontology Development.
105. PCI'09, The 13th Panhellenic Conference on Informatics.
106. ACIT'09, The International Arab Conference of Information Technology.
107. ORM'09, International Workshop on Fact-Oriented Modeling.
108. SWWS'09, 5th International IFIP Workshop on Semantic Web & Web Semantics
109. VORTE'09, International Workshop on Vocabularies, Ontologies and Rules for The Enterprise.
110. LOAIT'09, International Workshop on Legal Ontologies and Artificial Intelligence Techniques.
111. KARE'09, 2nd International Workshop on Knowledge Acquisition, Reuse and Evaluation.
112. ESAS'09, 4th IEEE International Workshop on Engineering Semantic Agent Systems.
113. ICIS'08, International Conference on Information Systems.
114. FLOW'08, International Workshop on Fuzzy Logic On the Web.
115. CBMS'08, The 21 IEEE International Symposium on Computer-Based Medical Systems.
116. ORM'08, Workshop on Object-Role Modeling.
117. ICDE'08, The 25th IEEE International Conference on Data Engineering (Demo track, Reviewer)
118. WOMSDE'08, Workshop on Ontologies and Metamodeling in Software and Data Engineering .
119. VORTE'08, EDOC International Workshop on Vocabularies, Ontologies and Rules for The Enterprise.
120. SWAE'2008, The 1st International Workshop on Semantic Web Architectures For Enterprises.
121. ESAS'08, The 3rd IEEE International Workshop on Engineering Semantic Agent Systems.
122. SWWS'08, 5th International IFIP Workshop on Semantic Web & Web Semantics
123. ESWC'07, The 4th European Semantic Web Conference.
124. ESAS'07, International Workshop on Engineering Semantic Agent.
125. CBMS'07, The 20th IEEE International Symposium on Computer-Based Medical Systems.
126. SWAE'2007, The 1st International Workshop on Semantic Web Architectures For Enterprises.
127. BADE 2007, IEEE Workshop on Biomedical Applications for Digital Ecosystems.
128. ORM'7, Workshop on Object-Role Modeling.
129. VORTE'07, EDOC International Workshop on Vocabularies, Ontologies and Rules for The Enterprise.
130. SWWS'07, IFIP WG 2.12 & WG 12.4 Workshop on Web Semantics.
131. SAMI'07, the Third IEEE Workshop on Situation Management.
132. LOAIT'07, International Workshop on Legal Ontologies and Artificial Intelligence Techniques.
133. ICDM'06, The IEEE Workshop on Data Mining in Bioinformatics.
134. CBMS'06, The 19th IEEE International Symposium on Computer-Based Medical Systems.
135. WOMSDE'06, The 1st Workshop on Ontologies and Metamodeling in Software and Data Engineering.
136. ORM'6, Workshop on Object-Role Modeling.
137. SWWS'06, IFIP WG 2.12 & WG 12.4 Workshop on Web Semantics.
138. SWIG06, The Canadian Semantic Web Working Symposium 2006.
139. DMB'06, IEEE ICDM 2006 Workshop on Data Mining in Bioinformatics.
140. VORTE'06, EDOC International Workshop on Vocabularies, Ontologies and Rules for The Enterprise.
141. ESAS'06, International Workshop on Engineering Semantic Agent.
142. ESWC'06, Industry Forum.
143. ODBase'05, International Conference on Ontologies, Databases and Applications of Semantics.
144. ORM'5, Workshop on Object-Role Modeling.
145. SWWS'05, IFIP WG 2.12 & WG 12.4 Workshop on Web Semantics.
146. VORTE'05, EDOC International Workshop on Vocabularies, Ontologies and Rules for The Enterprise.
147. LOAIT'05, International Workshop on Legal Ontologies and Artificial Intelligence Techniques.
148. SWCASE'05 Semantic Web Case Studies and Best Practices for eBusiness.
149. ESWC'05 Industry Forum.
150. IKE, The 2004 International Conference on Information and Knowledge Engineering.
151. SWEB 2004, International Workshop on Semantic Web Technologies in Electronic Business.

Online Courses

Slides and video-recorded lectures (YouTube Channel)

1. **Natural Language Processing** (COMP9312), PhD Program in Computer Science.
2. **Human Rights and IT** (LAIT6330), Master in Law and Information Technology.
3. **Artificial Intelligence** (COMP338, ENCS434), Computer Science and Computer Engineering programs.
4. **Advanced Artificial Intelligence** (SCOM7341), Master Program in Computing.
5. **Knowledge Engineering** (SCOM7348, SWEN7310), Master programs in Computing, and in Software Engineering.
6. **Data Engineering and Big Data Technologies** (SWEN739,COMP9316), Master Program in Software Engineering, and PhD Program in Computer Science.
7. **Web Data Management** (MCOM7348), Master Program in Computing.
8. **Data & Business Process Modelling** (COMP438), Computer Science Program.
9. **Philosophy of Language** (PHIL4381), University general elective.
10. **Discrete Mathematics** (COMP233), Computer Science Program.
11. **Artificial Intelligence and Machine Learning** (ASDS7381), Master Program of Applied Statistics and Data Science.
12. **Process and Data Modeling** (Tutorial), to IT middle-management public servants.
13. **Data Integration and Open Information System** (Tutorial), to IT middle-management public servants.
14. **Ontology Engineering and Lexical Semantics** (Tutorial), to IT middle-management public servants.
15. **Database Systems** (COMP333), Computer Science Program.
16. **Introduction to Computer** (COMP121), University level for non-Science/Engineering/IT students.
17. **Computer and Programing** (COMP133/COMP142), Computer Science & Computer Engineering programs.

Students

PhD Students:

1. Supervisor: Alaa Aljabari: Relationship Extraction. Birzeit University. (20232 - On)
2. Supervisor: Sanad Malaysha: Word Sense disambiguation. Birzeit University. (2022 - On)
3. Supervisor: Diyam Akra: Morphological Tagging. Birzeit University. (2022 - On)
4. Supervisor: Nagham Hamad: Domain Adaptation and Adversarial Learning. Birzeit University. (2022 - On)
5. Co-Supervisor: Eman Karajah: Arabic Word Substitution. Palestinian Polytechnic University. (2022 - On)
6. Examiner: Abdellah El Mikki: Cross-Dialectal Adaptation in Arabic NLP: Leveraging Modern Standard Arabic for Enhanced Performance in Dialectal Variants . Mohammed VI Polytechnic University, Morocco. (2023)
7. Co-Supervisor: Mamoun AbuHellu: Cross-Lingual Mapping of Lexical Ontologies with Automatic Translation. Birzeit University & Milano-Beccoca University. (2012 - 2016)
8. Examiner: Fuad Alattar, Artificial Intelligence Frameworks for Sentiment Variations' Reasoning and Emerging Topic Detection, The British University in Dubai, UAE. 2021
9. Examiner: Souheila Ben Guirat, Hybrid Approaches for Arabic Information Retrieval, Manouba University, Tunisia. 2020
10. Examiner: Manar Alkhatib, Neural Machine Translation for Arabic Language, The British University in Dubai, UAE. 2019
11. Examiner: Shastri Lakshman Nimmagadda, Ontology Based Data Warehousing for Mining of Heterogeneous and Multidimensional Data Sources, Curtin University, Australia.
12. Examiner: Ingrid Pappel, Paperless: Management as a Foundation for the Application of e-Governance in Local Governments. Tallinn University Of Technology, Estonia.
13. Examiner: Nazir Ahmad Sharif: Specification for an Ontology Server Assisting a Player to Commit to an Ontology for Interoperating Information Systems. University of Queensland. Australia. 2009

Masters Students (selected):

1. Supervisor(MS): Tymaa Hammouda: Software decisions for enhancing the performance of NLP systems. Birzeit University.
2. Supervisor(MS): Kamal Jarrar: Part-of-Speech Tagging for Palestinian Dialect Using Bi-LSTM.

3. Supervisor(MS): Nour Shayeb: Performance Analysis of Machine Learning Algorithms in predicting missing values in the Birzeit University Arabic Lexicographic Database. Birzeit University. 2022
4. Supervisor(MS): Ala Omar: Arabic Named Entity Disambiguation and Linking in Software Engineering. Birzeit University. 2022
5. Supervisor(MS): Sana Ghanam: Arabic Synonyms Extraction. Birzeit University. 2021
6. Supervisor(MS): Qutaiba Mustafa: Detecting and Classifying Software Requirement in App Stores. Birzeit University. 2021
7. Supervisor(MS): Manar Awadallah: Predicting the Word Number in Arabic. Birzeit University. 2021
8. Co-Supervisor(MS): Shurooq Morrar: Social Media Mobilization in Time of Crisis: Palestinian Employment in Lebanon. Birzeit University. 2020
9. Supervisor(MS): Diana Al-Hafi: User Experience Evaluation for Birzeit Linguistic Search Engine. Birzeit University. 2019
10. Examiner(MS): Abdulfattah Safa: Enhancing Quality Arabic Web Content through Multilingual Information Retrieval. Birzeit University. 2018
11. Examiner(MS): Mohamad Ateeq: Speech Based Call System to Evaluate the Meaning and Grammar Errors in English. Birzeit University. 2018
12. Examiner(MS): Abd-aljabar Huroob: Using Computational Linguistics Applications in Arabic Language - Vocalizer and Summarizer as Models. Birzeit University. 2018
13. Supervisor(MS): Eman Karajah: **Towards Building an Arabic Thesaurus Automatically**. Birzeit University. 2015
14. Supervisor(MS): Diyam Akra: **Towards Building a Corpus for Palestinian Dialect**. Birzeit University. 2015
15. Examiner(MS): Mahmoud El-Askary : An Ontology-Based Approach for Diagnosing Date Palm Diseases. Islamic University of Gaza. 2015
16. Examiner(MS): Dua Awwad: Performance Analysis for Business Process Modeling Notation, Palestinian Polytechnic University. 2014
17. Supervisor(MS): Rana Rishmawi: Towards Building an Arabic Upper-Level Ontology. Birzeit University. 2013
18. Examiner(MS): Yahya Dalluol: An Ontology-Based Approach to Support the Process of Judging Hadith Isnad. Islamic University of Gaza. 2013
19. Examiner(MS): Rana Abdo: Ontology-Based Digital Libraries, AlQuds University, 2014
20. Supervisor(MS): Mohammad M. Melhem: Advanced Matching Function Towards Linking Arabic Ontology with the English WordNet. Birzeit University. 2013
21. Supervisor(MS): Rami Kalil: **Mapping ORM into OWL2 and Back**. Birzeit University. 2012
22. Co-Supervisor(MS): Georgiou M: Semantic Web Pipes. Cyprus University. 2009
23. Co-Supervisor(MS): Savvides C: Semantic Data Mashups. Cyprus University. 2009
24. Supervisor(MS): Al-Damag M: Mapping ORM into DIG. Vrije Universiteit Brussel. 2006
25. Supervisor(BA): Deik A, Farraj B, Hawash A: Querying the Data Web. Birzeit University. 2009
26. Supervisor(BA): Al-Badawi M, Aish S, Samara H: Towards Building an Arabic Ontology. Birzeit University. 2009
27. Supervisor(BA): Manoli A: Semantic Mashups at the Browser Side. Cyprus University. 2009
28. Supervisor(BA): Gielen G: Gloss Support in Ontology Modeling. Vrije Universiteit Brussel. 2007
29. Supervisor(MS): Ugurel A: Ontology-driven Web Forms. Vrije Universiteit Brussel. 2007
30. Supervisor(MS): Yan Gu: Incorporating SUMO into DOGMA. Vrije Universiteit Brussel. 2007
31. Supervisor(MS): Hoang H: Verbalization of Ontologies. Vrije Universiteit Brussel. 2005
32. Supervisor(MS): Murhaf M: Generation of RDF from Databases. Vrije Universiteit Brussel. 2004
33. Supervisor(BA): Demey J: Realizing Ontological Commitments. Vrije Universiteit Brussel. 2003
34. Supervisor(MS): Lisovoy A: DogmaModeler: Ontology Modeling. Vrije Universiteit Brussel. 2003
35. Supervisor(MS): W. Wang: Ontologization of WordNet. Vrije Universiteit Brussel. 2002
36. Supervisor(MS): M. Orzu: Ontology Mining from Databases. Vrije Universiteit Brussel. 2002
37. Supervisor(BA): P. Segers: Semantic-Based Digital Libraries. Vrije Universiteit Brussel. 2001

Articles

Google Scholar: Citations: 4500, h-index: 36

2024

- Mustafa Jarrar, Tymaa Hammouda: **Qabas: An Open-Source Arabic Lexicographic Database**. In Proceedings of the International Conference on Language Resources and Evaluation (LREC-COLING 2024), Torino, Italy. 2024.
- Sanad Malaysha, Mustafa Jarrar, Mohammed Khalilia: **NLU-STR at SemEval-2024 Task 1: Generative-based Augmentation and Encoder-based Scoring for Semantic Textual Relatedness** In Proceedings of the SemEval 2024 Shared Task 1 (Semantic Relatedness). 2024.

2023

- Mustafa Jarrar, Sanad Malaysha, Tymaa Hammouda, Mohammad Khalilia: **SALMA: Arabic Sense-Annotated Corpus and WSD Benchmarks**. In Proceedings of the 1st Arabic Natural Language Processing Conference (ArabicNLP), Part of the EMNLP 2023. ACL.
- Amal Nayouf, Tymaa Hammouda, Mustafa Jarrar, Fadi zaraket, Mohamad-Bassam Kurdy: **Nâbra: Syrian Arabic Dialects with Morphological Annotations**. In Proceedings of the 1st Arabic Natural Language Processing Conference (ArabicNLP), Part of the EMNLP 2023. ACL.
- Mustafa Jarrar, Ahmet Birim, Mohammed Khalilia, Mustafa Erden, and Sana Ghanem: **ArBanking77: Intent Detection Neural Model and a New Dataset in Modern and Dialectical Arabic**. In Proceedings of the 1st Arabic Natural Language Processing Conference (ArabicNLP), Part of the EMNLP 2023. ACL.
- Haneen Liqreina, Mustafa Jarrar, Mohammed Khalilia, Ahmed Oumar El-Shangiti, Muhammad AbdulMageed: **Arabic Fine-Grained Entity Eecognition**. In Proceedings of the 1st Arabic Natural Language Processing Conference (ArabicNLP), Part of the EMNLP 2023. ACL.
- Mustafa Jarrar, Muhammad Abdul-Mageed, Mohammed Khalilia, Bashar Talafha, AbdelRahim El-madany, Nagham Hamad, Alaa' Omar: **WojoodNER 2023: The First Arabic Named Entity Recognition Shared Task**. In Proceedings of the 1st Arabic Natural Language Processing Conference (Arabic- NLP), Part of the EMNLP 2023. ACL.
- Nouran Khallaf, Elin Arfon, Mo El-Haj, Jon Morris, Dawn Knight, Paul Rayson, Tymaa Hammouda, Mustafa Jarrar: **Open-source thesaurus development for under-resourced languages: a Welsh case study**. The 4th LDK Conference on Language, Data and Knowledge, Vienna, Austria, 12-15 September 2023
- Nagham Hamad, Mustafa Jarrar, Mohammad Khalilia, Nadim Nashif: **Offensive Hebrew Corpus and Detection using BERT**. The 20th ACS/IEEE International Conference on Computer Systems and Applications (AICCSA). IEEE. Egypt. 2023
- Sana Ghanem, Mustafa Jarrar, Radi Jarrar, Ibrahim Bounhas: **A Benchmark and Scoring Algorithm for Enriching Arabic Synonyms**. The 12th International Global Wordnet Conference (GWC2023), Global Wordnet Association. (pp.274-283). San Sebastian, Spain, 2023
- Sanad Malaysha, Mustafa Jarrar, Mohammad Khalilia: **Context-Gloss Augmentation for Improving Arabic Target Sense Verification**. The 12th International Global Wordnet Conference (GWC2023), Global Wordnet Association. (pp. 254-262). San Sebastian, Spain, 2023

2022

- Mustafa Jarrar, Mohammed Khalilia, Sana Ghanem: **Wojood: Nested Arabic Named Entity Corpus and Recognition using BERT**. In Proceedings of the International Conference on Language Resources and Evaluation (LREC 2022), Marseille, France. 2022
- Mustafa Jarrar, Fadi Zaraket, Tymaa Hammouda, Daanish Masood, Martin Waehlsch: **Lisan: Yemeni, Iraqi, Libyan, and Sudanese Arabic Dialect Corpora with Morphological Annotations**. The 20th ACS/IEEE International Conference on

Computer Systems and Applications (AICCSA). Pages(-). IEEE. Egypt. 2023 arXiv, DOI 10.48550/ARXIV.2212.06468. 2023

- Karim El Haff, Mustafa Jarrar, Tymaa Hammouda, Fadi Zaraket: **Curras + Baladi: Towards a Levantine Corpus**. In Proceedings of the International Conference on Language Resources and Evaluation (LREC 2022), Marseille, France. 2022

2021

- Mustafa Jarrar: **The Arabic Ontology - An Arabic Wordnet with Ontologically Clean Content**. Applied Ontology Journal, 16:1, 1-26. IOS Press. 2021
- Moustafa Al-Hajj, Mustafa Jarrar: **ArabGlossBERT: Fine-Tuning BERT on Context-Gloss Pairs for WSD**. In Proceedings of the International Conference on Recent Advances in Natural Language Processing (RANLP 2021). PP 40--48, 2021
- Moustafa Al-Hajj, Mustafa Jarrar: **LU-BZU at SemEval-2021 Task 2: Word2Vec and Lemma2Vec performance in Arabic Word-in-Context disambiguation**. In Proceedings of the Fifteenth Workshop on Semantic Evaluation (SemEval2021) Task 2: Multilingual and Cross-lingual Word-in-Context Disambiguation (MCL-WiC). PP 748--755, Association for Computational Linguistics. 2021
- Eman Naser-Karajah, Nabil Arman, Mustafa Jarrar: **Current Trends and Approaches in Synonyms Extraction: Potential Adaptation to Arabic**. In Proceedings of the 2021 International Conference on Information Technology (ICIT). PP 748--755, Association for Computational Linguistics. pp. 428-434, IEEE. 2021
- Mustafa Jarrar, Eman Karajah, Muhammad Khalifa, Khaled Shaalan: **Extracting Synonyms from Bilingual Dictionaries**. The 11th International Global Wordnet Conference (GWC2021), Global Wordnet Association. (pp. 215-222). Pretoria, South Africa, 2021
- Kareem Darwish, Nizar Habash, Mourad Abbas, Hend Al-Khalifa, Huseein T. Al-Natsheh, Houda Bouamor, Karim Bouzoubaa, Violetta Cavalli-Sforza, Samhaa R. El-Beltagy, Wassim El-Hajj, Mustafa Jarrar, Hamdy Mubarak: **A Panoramic Survey of Natural Language Processing in the Arab World**. Communications of the ACM, April 2021, Vol. 64 No. 4, Pages 72-81
- Mustafa Jarrar: **Digitization of Arabic Lexicons**. Arabic Language Status Report. UAE Ministry of Culture and Youth. Pages 214-2017. Dec 2020

2019

- Mustafa Jarrar, Hamzeh Amayreh: **An Arabic-Multilingual Database with a Lexicographic Search Engine**. The 24th International Conference on Applications of Natural Language to Information Systems (NLDB 2019). Pages(234-246). LNCS 11608, Springer. 2019
- Mustafa Jarrar, Hamzeh Amayreh, John P. McCrae: **Representing Arabic Lexicons in Lemon - a Preliminary Study**. The 2nd Conference on Language, Data and Knowledge (LDK 2019). Pages(29-33). CEUR, Volume 2402. ISSN:1613-0073. Leipzig, Germany. 2019
- Diana Alhafi, Anton Deik, Mustafa Jarrar: **Usability Evaluation of Lexicographic e-Services**. The 16th IEEE/ACS International Conference on Computer Systems and Applications (AICCSA). Pages(1-7). IEEE. Abu Dhabi, UAE. 2019

2018

- Mustafa Jarrar, Fadi Zaraket, Rami Asia, Hamzeh Amayreh: **Diacritic-Based Matching of Arabic Words**. ACM Asian and Low-Resource Language Information Processing. Volume 18, No 2, Pages(10:1-10:21), ACM, ISSN:2375-4699. December, 2018
- Paolo Ceravolo, Antonia Azzini, Marco Angelini, Tiziana Catarci, Philippe Cudre-Mauroux, Ernesto Damiani, Alexandra Mazak, Maurice Van Keulen, Mustafa Jarrar, Giuseppe Santucci, Kai-Uwe Sattler, Monica Scannapieco, Manuel Wimmer, Robert Wrembel, Fadi Zaraket: **Big Data Semantics**. Journal on Data Semantics. Volume 7, Issue 2, Pages(65-85). Springer. May, 2018

- Mustafa Jarrar: **Search Engine for Arabic Lexicons**. The 5th Conference on Translation and the Problematics of Cross-cultural Understanding. The Forum for Arab and International Relations. Doha, Qatar. December, 2018

2017

- Mustafa Jarrar, Werner Ceusters: **Classifying Processes and Basic Formal Ontology**. The 8th International Conference on Biomedical Ontology (ICBO), Newcastle, UK. September, 2017
- Diab Abuaiadah, Dileep Rajendran, Mustafa Jarrar: **Clustering Arabic Tweets for Sentiment Analysis**. The 2017 IEEE/ACS 14th International Conference on Computer Systems and Applications. Pages(499-506). IEEE Computer Society. ISBN:9781538635810. (doi.10.1109/AICCSA.2017.162). Hammamet, Tunisia. 2017
- Mustafa Jarrar, Nizar Habash, Faeq Alrimawi, Diyam Akra, Nasser Zalmout: **Curras: An Annotated Corpus for the Palestinian Arabic Dialect**. Journal Language Resources and Evaluation. Pages(745-775). Volume(51), Issue(3). Springer (doi.org/10.1007/s10579-016-9370-7). 2017

2016

- Mamoun Abu Helou, Matteo Palmonari, Mustafa Jarrar: **Effectiveness of Automatic Translations for Cross-Lingual Ontology Mapping**. Journal of Artificial Intelligence Research, Special Track on Cross-language Algorithms and Applications. Pages(165-208). Volume(55), Number(1). AI Access Foundation. 2016

2015

- Mustafa Jarrar, Anton Deik: **The Graph Signature: A Scalable Query Optimization Index for RDF Graph Databases Using Bisimulation and Trace Equivalence Summarization**. International Journal on Semantic Web and Information Systems, 11(2), Pages(36-65). IGI Global. DOI:10.4018/IJSWIS.2015040102. April-June, 2015

2014

- Mustafa Jarrar, Nizar Habash, Diyam Akra, Nasser Zalmout: **Building a Corpus for Palestinian Arabic: a Preliminary Study**. Arabic Natural Language Processing Workshop, at the Conference on Empirical Methods in Natural Language Processing (EMNLP 2014). Pages(18-27). Association for Computational Linguistics. ISBN:9781937284961. Qatar. October, 2014
- Mamoun Abu Helou, Matteo Palmonari, Mustafa Jarrar, Christiane Fellbaum: **Towards Building Lexical Ontology via Cross-Language Matching**. The 7th Conference on Global WordNet. Pages(346-354). Global WordNet Association. ISBN:7492329949978. Tartu, Estonia. January, 2014
- Antonio Lucas Soares, Carla Sofia Pereira, Mustafa Jarrar (eds): **Proceedings of the International Workshop on Ontology Content and Evaluation (OnToContent 2014)**. In OTM 2014 Workshops. Page 575. LNCS 8842, Springer. ISBN:9783662455494. October, 2014

2012

- Mustafa Jarrar and Marios D. Dikaiakos: **A Query Formulation Language for the Data Web**. The IEEE Transactions on Knowledge and Data Engineering. Volume 24, Number 4, Pages(783-798). IEEE Computer Society. April, 2012
- Gianluca Elia, Mustafa Jarrar: **Guest Editorial: Knowledge Management and e-Human Resources Practices for Innovation**. The International Journal of Knowledge and Learning (IJKL). Pages(1-5). Volume (8), Number(1/2). Inderscience Publishers. 2012
- Mustafa Jarrar, Amanda Hicks, Matteo Palmonar (eds): **Proceedings of the International Workshop on Ontology Content and Evaluation (OnToContent 2012)**. In OTM 2012 Workshops. Page 419. LNCS 7567, Springer. ISBN:9783642336171. September, 2012
- Mustafa Jarrar: **Proceedings of the 1st Palestinian Conference on e-Governance and e-Services**. Sina Institute at Birzeit University. June, 2012

2011

- Mustafa Jarrar: **Building a Formal Arabic Ontology (Invited Paper)**. Proceedings of the Experts Meeting on Arabic Ontologies and Semantic Networks. ALECSO, Arab League. Tunisia. July, 2011
- Mustafa Jarrar, Anton Deik, Bilal Faraj: **Ontology-Based Data and Process Governance Framework -The Case of e-Government Interoperability in Palestine**. The IFIP International Symposium on Data-Driven Process Discovery and Analysis (SIMPDA). Pages(83-98). ISBN:9788890312021. Campione, Italy. June, 2011
- Paolo Ceravolo, Chengfei Liu, Mustafa Jarrar, Kai-Uwe Sattler: **Special Issue on Querying the Data Web -Novel techniques for querying structured data on the web**. The World Wide Web Journal. Volume(14), Issue (5-6). Springer. ISSN:1573-1413. August, 2011.

2010

- Mustafa Jarrar, Marios D. Dikaiakos: **Querying the Data Web: the MashQL Approach**. IEEE Internet Computing. Volume 14, No 3. Pages(58-670). IEEE Computer Society. ISSN:1089-7801. May, 2010
- Anton Deik, Bilal Faraj, Ala Hawash, Mustafa Jarrar: **Towards Query Optimization for the Data Web - Two Disk-Based algorithms: Trace Equivalence and Bisimilarity**. The International Conference on Intelligent Semantic Web – Applications and Services. Pages(17:1-7). ACM. ISBN:9781450304757. June, 2010
- Rami Hodrob, Mustafa Jarrar: **ORM to OWL 2 DL Mapping**. The International Conference on Intelligent Semantic Web – Applications and Services. Pages(9:1-7). ACM. ISBN:9781450304757. June, 2010
- Anton Deik, Bilal Faraj, Ala Hawash, Mustafa Jarrar: **Towards Query Optimization for the Data Web - Two Disk-Based algorithms: Trace Equivalence and Bisimilarity**. The 3rd Palestinian International Conference on Computer and Information Technology (PICCIT 2010). Hebron, Palestine. March, 2010
- Jamal Daher, Mustafa Jarrar: **Towards a Methodology for Building Ontologies – Classify by Properties (in Arabic)**. The 3rd Palestinian International Conference on Computer and Information Technology (PICCIT). Hebron, Palestine. March, 2010
- Mustafa Jarrar, Paolo Ceravolo: (eds): **Proceedings of the International Workshop on Ontology Content and Evaluation (OnToContent 2010)**. In OTM 2010 Workshops. Page 367. LNCS 6428, Springer. ISBN:9783642169601. October, 2010

2009

- Mustafa Jarrar and Marios D. Dikaiakos: **A Data Mashup Language for the Data Web**. Linked Data on the Web Workshop, at the International World-Wide Web Conference (WWW 2009). ACM. ISSN:1613-0073. April, 2009
- Sergey Lukichev, Mustafa Jarrar: **Graphical Notations for Rule Modeling**. Book Chapter in the "Handbook of Research on Emerging Rule-Based Languages and Technologies". Chapter 4. Pages(76-98). IGI Global. ISBN:1605664022. 2009

2008

- Mustafa Jarrar and Marios D. Dikaiakos: **MashQL: a Query-by-Diagram Topping SPARQL —Towards Semantic Data Mashups**. Ontologies and Information Systems for the Semantic Web Workshop, at the ACM International Conference on Information and Knowledge Management (CIKM 2008). Pages(89-96). ACM. ISBN:9781605582559. November, 2008
- Mustafa Jarrar: **Towards Effectiveness and Transparency in e-Business Transactions, An Ontology for Customer Complaint Management**. Book Chapter in "Semantic Web Methodologies for E-Business Applications". chapter 7. Pages(127-149). IGI Global. ISBN:9781605660660. 2008
- Mustafa Jarrar and Stijn Heymans: **Towards Pattern-Based Reasoning for Friendly Ontology Debugging**. Journal of Artificial Intelligence Tools. Volume 17. No 4. World Scientific Publishing. August, 2008
- Mustafa Jarrar and Robert Meersman: **Ontology Engineering —The DOGMA Approach**. Book Chapter in "Advances in

- Ramon F. Brena, Andreas Schmidt, Mustafa Jarrar, Werner Ceusters, Francisco J. Cantu: (eds): **Proceedings of the International Workshop on Ontology Content and Evaluation (OnToContent+QSI)**. In OTM 2008 Workshops. Page 583. LNCS 5333, Springer. ISBN:9783540888741. November, 2008

2007

- Mustafa Jarrar: **Mapping ORM into the SHOIN/OWL Description Logic —Towards a Methodological and Expressive Graphical Notation for Ontology Engineering**. OTM 2007 workshops: Proceedings of the International Workshop on Object-Role Modeling (ORM). Pages(729-741). LNCS 4805, Springer. ISBN:9783540768890. Portugal. November, 2007
- Mustafa Jarrar: **Towards Automated Reasoning on ORM Schemes —Mapping ORM into the DLR_idf Description Logic**. The 26th International Conference on Conceptual Modeling (ER). Pages(181-197). LNCS 801, Springer. ISBN:9783540755623. Auckland, New Zealand. November, 2007
- Mustafa Jarrar, Andreas Schmidt, Claude Ostin, and Werner Ceusters (eds): **Proceedings of the International Workshop on Ontology Content and Evaluation (OnToContent)**. On the Move to Meaningful Internet Systems: OTM 2007 Workshops. Page 509. LNCS 4805. Springer. ISBN:9783540768876. Albufeira, Portugal. November, 2007

2006

- Mustafa Jarrar: **Towards the Notion of Gloss, and the Adoption of Linguistic Resources in Formal Ontology Engineering**. The 15th International World Wide Web Conference (WWW2006). Pages(497-503). ACM Press. ISBN:1595933239. Edinburgh, Scotland. May, 2006
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- Mustafa Jarrar and Stijn Heymans: **Unsatisfiability Reasoning in ORM Conceptual Schemes**. Current Trends in Database Technology – The International Conference on Extending Database Technology (EDBT 2006). Pages(517-534). LNCS 4254, Springer. ISBN:3540467882. Munich, Germany. March, 2006
- Mustafa Jarrar, Claude Ostin, Werner Ceusters, Andreas Persidis (eds): **Proceedings of the International Workshop on Ontology Content and Evaluation (OnToContent 2006)**. On the Move to Meaningful Internet Systems 2006: OTM 2006 Workshops. Page 1011. LNCS 4278, Springer. ISBN:9783540482734. Montpellier, France. November, 2006
- Katia Sycara, Elizabeth Chang, Ernesto Damiani, Mustafa Jarrar, and Tharam Dillon (eds): **Proceedings of the 2nd IFIP WG 2.12 and WG 12.4 International Workshop on Web Semantics (SWWS'06)**. On the Move to Meaningful Internet Systems 2006: OTM 2006 Workshops. Page 1723. LNCS 4278, Springer. ISBN:9783540482734. Montpellier, France. November, 2006
- Mustafa Jarrar, Maria Keet, and Paolo Dongilli: **Multilingual Verbalization of ORM Conceptual Models and Axiomatized Ontologies**. Technical report. STARLab, Vrije Universiteit Brussel. February, 2006

2005

- Mustafa Jarrar: **Modularization and Automatic Composition of Object-Role Modeling (ORM) Schemes**. The International Workshop on Object-Role Modeling (ORM). OTM2005 Workshops. Pages(613-625). LNCS 3762, Springer. ISBN:3540297391. Larnaca, Cyprus. November, 2005
- Mustafa Jarrar: **Towards Methodological Principles for Ontology Engineering**. PhD Thesis. Vrije Universiteit Brussel. May, 2005
- Tharam Dillon, Ling Feng, Mustafa Jarrar, Aldo Gangemi, Joost Breuker, Jos Lehmann, and Andre Valente (eds): **The 1st IFIP WG 2.12 and WG 12.4 International Workshop on Web Semantics (SWWS'05)**. On the Move to Meaningful

Internet Systems 2005: OTM 2005 Workshops. Page 808. LNCS 3762. Springer. ISBN:9783540297390. Larnaca, Cyprus. November, 2005

- Robert Meersman, Zahir Tari, Pilar Herrero, Gonzalo Mendez, Lawrence Cavedon, David Martin, Annika Hinze, George Buchanan, Maria S. Perez, Victor Robles, Jan Humble, Antonia Albani, Jan L. G. Dietz, Herve Panetto, Monica Scannapieco, Terry A. Halpin, Peter Spyns, Johannes Maria Zaha, Esteban Zimanyi, Emmanuel Stefanakis, Tharam S. Dillon, Ling Feng, Mustafa Jarrar, Jos Lehmann, Aldo de Moor, Erik Duval, Lora Aroyo: (eds): **On the Move to Meaningful Internet Systems 2005: OTM 2005 Workshops**. LNCS 3762, Springer. ISBN:3540297391. Larnaca, Cyprus. November, 2005

2004

- Karl Aberer, Tiziana Catarci, Philippe Cudre-Mauroux, Tharam Dillon, Stephan Grimm, Mohand-Said Hacid, Arantza Illarramendi, Mustafa Jarrar, Vipul Kashyap, Massimo Mecella, Eduardo Mena, Erich Neuhold, Aris Ouksel, Thomas Risse, Monica Scannapieco, Felix Saltor, Luca De Santis, Stefano Spaccapietra, Steffen Staab, Rudi Studer, Olga De Troyer: **Emergent Semantics Systems**. Semantics of a Networked World - Semantics for Grid Databases (ICSNW2014). Pages(14-44). LNCS 3226, Springer. ISBN:3540236090. Paris, France. June, 2004
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- Mustafa Jarrar and Aldo Gangemi (eds): **Proceedings of the 2nd International Workshop on Regulatory Ontologies (WORM 2004)**. On the Move to Meaningful Internet Systems 2004: OTM 2004 Workshops. Page 568. LNCS 3292, Springer. ISBN:9783540236641. Larnaca, Cyprus. November, 2004

2003

- Mustafa Jarrar, Jan Demey, and Robert Meersman: **On Using Conceptual Data Modeling for Ontology Engineering**. Journal on Data Semantics, Special issue on "Best papers from the ER/ODBASE/COOPIS 2002 Conferences". Pages(185-207). LNCS 2800. No 1. Springer. ISBN:3540204075. October, 2003
- Mustafa Jarrar, Ruben Verlinden, and Robert Meersman: **Ontology-Based Customer Complaint Management**. OTM 2003 Workshops, Proceedings of the 1st International Workshop on Regulatory Ontologies and the Modeling of Complaint Regulations. Pages(594-606). LNCS 2889, Springer. ISBN:9783540204947. Sicily, Italy. November, 2003
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Online Tools and Databases

with scientifically-original features and wide use.

- **Arabic-Multilingual Lexicographic Database (with Search Engine)**

- The largest lexicographic database ever for Arabic.
- Includes 150 Arabic lexicons, digitized over 10 years.
- The first of its kind (e.g., no similar tools for English lexicons).
- Search 150 lexicons for definitions, synonyms, and translations.
- Free access to people with accessible APIs for NLP applications.
- Awarded *Mohammed Bin Rashid Award* for the Arabic Language.

- **The Arabic Ontology**

- An Arabic Wordnet with ontologically-clean content.
- Classification of the meanings of the Arabic terms, based on state-of-the-art science.
- Semantically linked with Arabic lexicons, WordNet, WikiData, and others.
- The first linguistic ontology of its kind.
- Open source, online, and accesable via API.

- **Currasat - Arabic Dialect Corpora**

- Morphological-annotated corpus (about 1.35K words) for seven dialects
- Palestinian, Lebanese, Syrian, Iraqi, Libyan, Yemeni, and Sudanese.
- Each word is annotated with 16 features.
- To enable machines to understand Arabic dialect.
- Open source, and heavily used by non-native Arabic speakers.

- **Wojood NER (Nest Named-Entities Corpus and Model)**

- About 550K tokens (MSA and dialect) that are manually annotated.
- 21 entity types (e.g., person, organization, location, event, date, etc)..
- Contains about 75K entities and 22.5% of which are nested.
- Covers multiple domains and was annotated with nested entities.
- F1-score 88.4%.
- Open source, online, and accesable via API.

- **Tools for Natural Language Understanding**

- A set of open source tools for Arabic NLP.
- Lemmatizer, word-sense disambiguation, etc.
- Jaccard Function - union, intersection and similarity between Arabic words.
- Diacritic-Based Matching of Arabic words.
- A framework and 30 parsers for digitizing lexicons.

- **Zinnar**

- The Palestinian e-government interoperability framework.
- Officially endorsed by the Palestinian Cabinet, and used by all ministries.
- Six years of voluntary work to help the government.

- Open source and online.
- **MashQL**
 - A query formulation language for semi-structured data.
 - A novel approach for querying the Data Web, graph databases, and data mashups.
 - Its indexing is way faster than Oracle graph database.
 - Open source and online.
- **DogmaModeler**
 - Open source ontology modeling tool based on ORM.
 - Supports: reasoning in Description Logic, verbalization, and modularization.
 - Widely used by the research community, 300 downloads a month.
- **ORM Markup language 3**
 - Open source XML-based serialization for ORM Business rules.
 - A conceptual markup language for the ORM graphical notation.
 - Supports modular schemes, and a decent set of metadata elements.
- **ORM verbalizer**
 - Open source multilingual verbalization for ORM in 11 languages.
 - Used for ontologies, conceptual models, business rules, and logic.
 - Arabic, English, Catalan, Spanish, Dutch, French, German, Italian, Lithuanian, Russian.
- **CContology**
 - Open source ontology for customer complaint management.
 - Supports 11 languages.

Technical Reports

2019

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- Rayan Dankar, Hadi Ghattas, Fadi Zaraket, Mustafa Jarrar, Hamzeh Amayreh, Walid Magdi, Alex Rutherford, Daanish Masood, Martin Waehlich: **Social Media Opinion Mining and Sentiment Analysis - Bahrain Elections**. Technical Report. United Nations. New York. 2018
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- Mustafa Jarrar, Faeq Alrimawi: **Statistics and Inter-Annotator Agreement Calculations of the Palestinian Dialect Corpus —Curras**. Technical Report. Birzeit University, Palestine. August, 2015
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- Mustafa Jarrar, Luk Vervenne, Diana Maynard: **HR-Semantics Roadmap- The Semantic Challenges and Opportunities in the Human Resources Domain**. Technical Report. The Ontology Outreach Advisory, Belgium. (OOA-HR/2007-08-20/v025). August, 2007
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- Asuncion Gomez-Perez, Elena Paslaru Bontas, Jens Hartmann, Lyndon Nixon, Mustafa Jarrar, Maria Carmen Suarez-Figueroa, and Raul Garcia-Castro: **Identification of Standards on Metadata for Ontologies**. Deliverable D1.3.2 (WP1.1), The Knowledge Web Network of Excellence (NoE) IST-2004-507482, Luxemburg. January, 2005
- Jerome Euzenat, Thanh Le Bach, Jesus Barrasa, Paolo Bouquet, Jan De Bo, Rose Dieng, Marc Ehrig, Manfred Hauswirth, Mustafa Jarrar, Ruben Lara, Diana Maynard, Amedeo Napoli, Giorgos Stamou, Heiner Stuckenschmidt, Pavel Shvaiko, Sergio Tessaris, Sven Van Acker, Ilya Zaihrayeu: **Ontology Alignment - State of the Art**. Deliverable D2.2.3 (WP2.2), The Knowledge Web Network of Excellence (NoE) IST-2004-507482, Luxemburg. January, 2005
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- Mustafa Jarrar, Jan Demey, Robert Meersman: **Exchanging ORM Schemas using a Conceptual Markup Language**. Technical Report. STAR Lab, Vrije Universiteit Brussel, Belgium. 2002
- Robert Meersman and Mustafa Jarrar: **An Architecture and Toolset for Practical Ontology Engineering**. Technical Report. STAR Lab, Vrije Universiteit Brussel, Belgium. 2002

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Talks and Featured Keynotes

1. **SALMA: Arabic Sense-Annotated Corpus and WSD Benchmarks** ([Video](#) - [Article](#) - [slides](#)). *Research Talk*. The First Arabic Natural Language Processing Conference (ArabicNLP 2023) - part of EMNLP-2023, Singapore. 7/12/2023
2. **Nabra: Syrian Arabic Dialects with Morphological Annotations** ([Video](#) - [Article](#) - [slides](#)). *Research Talk*. The First Arabic Natural Language Processing Conference (ArabicNLP 2023) - part of EMNLP-2023, Singapore. 7/12/2023
3. **ArBanking77: Intent Detection Neural Model and a New Dataset in Modern and Dialectical Arabic** ([Video](#) - [Article](#) - [slides](#) - [Poster](#)). *Research Talk*. The First Arabic Natural Language Processing Conference (ArabicNLP 2023) - part of EMNLP-2023, Singapore. 7/12/2023
4. **Arabic Fine-Grained Entity Eecognition** ([Video](#) - [Article](#) - [slides](#)). *Research Talk*. The First Arabic Natural Language Processing Conference (ArabicNLP 2023) - part of EMNLP-2023, Singapore. 7/12/2023
5. **WojoodNER 2023: The First Arabic Named Entity Recognition Shared Task** ([Video](#) - [Article](#) - [slides](#) - [Poster](#)). *Research Talk*. The First Arabic Natural Language Processing Conference (ArabicNLP 2023) - part of EMNLP-2023, Singapore. 7/12/2023
6. **Lisan: Yemeni, Iraqi, Libyan, and Sudanese Arabic Dialect Corpora with Morphological Annotations** ([Article](#) - [slides](#)). *Research Talk*. The 20th ACS/IEEE International Conference on Computer Systems and Applications (AICCSA), Cairo. 4/12/2023
7. **Offensive Hebrew Corpus and Detection using BERT** ([Article](#) - [slides](#)). *Research Talk*. The 20th ACS/IEEE International Conference on Computer Systems and Applications (AICCSA), Cairo. 4/12/2023
8. **Arabic Language Understanding Tools** ([slides](#)). *Featured Keynote*. UM6P NLP Day, Morocco. 28/11/2023
9. **Academic Integrity and Publication Challenges** ([video](#) - [Facebook](#) - [AlHadath](#)). *Chair*. Birzeit University Seminars, 2023
10. **On ChatGPT and Larage Language Models** ([video](#) - [slides](#)). *Invited Talk*. Birzeit University Seminars - ChatGPT in Academia, 2023
11. **A Benchmark and Scoring Algorithm for Enriching Arabic Synonyms** ([Article](#) - [slides](#)). *Research Talk*. The 12th International Global Wordnet Conference (GWC2023), San Sebastian, Spain, 2023
12. **Context-Gloss Augmentation for Improving Arabic Target Sense Verification.** ([Article](#) - [slides](#)). *Research Talk*. The 12th International Global Wordnet Conference (GWC2023), San Sebastian, Spain, 2023
13. **Arabic Natural Language Understanding as a Sustainable AI Research** ([Program](#) - [slides](#) - [video](#)). *Invited Talk*. The e-AGE21 international Confernce, Online, 14/12/2021
14. **Building Lexicographic Database for MSA and dialects** ([Conference](#) - [slides](#)). *Featured Keynote*. The e-AGE21 international Conference, Online, 28/9/2021
15. **Linguistic Ontologies for Knowledge Graphs** ([slides](#)). *Invited Talk*. FIP WG 2.6 Database Seminars, 21/1/2021
16. **Extracting Synonyms from Bilingual Dictionaries** ([Article](#) - [slides](#) - [video](#)). *Research Talk*. The 11th Global Wordnet Conference. South Africa 21/1/2021
17. **Linguistic Ontologies and Wordnets** ([slides](#) - [video](#)). *Invited Speaker*. The 11th Global Wordnet Conference. South Africa 18/1/2021
18. **Building the Largest Arabic-Multilingual Lexicographic Database** ([media](#) - [Lecture](#) - [video](#)). *Featured Keynote*. Shoman Cultural Forum - Abdul Hameed Shoman Foundation. Amman, Jordan 10/3/2020
19. **Computer Science Research under Israeli Occupation -my Story** ([Program](#)). *Keynote Speaker*. The 3rd International Meeting for Science in Palestine. Massachusetts Institute of Technology (MIT). Boston, USA 11/1/2020

20. **NLP Tools and Resources for Understanding Arabic Content** ([news](#)). *Keynote Speaker*. Understanding Internet Content Symposium - Birzeit University. Palestine 24/7/2019
21. **Social Media Analysis for Peace** *Invited Talk*. The United Nations AI for Peace and Security Workshop, Alan Turing Institute. London, UK 24/7/2019
22. **An Arabic-Multilingual Database with a Lexicographic Search Engine** ([slides](#)). *Research Talk*. The 24th International Conference on Applications of Natural Language to Information Systems (NLDB). Manchester, UK 26/6/2019
23. **Representing Arabic Lexicons in Lemon - a Preliminary Study** ([poster](#) - [article](#)). *Research Poster*. The 2nd Conference on Language, Data and Knowledge (LDK). Leipzig, Germany 23/5/2019.
24. **Big Linguistic Data Graph** ([slides](#) - [video](#)). *Invited Talk*. The First Arabic Speech Workshop. Qatar 29/4/2019
25. **Innovative Lexicographic Search Engine** ([program](#)). *Lightning Talk*. The 2019 NYUAD International Hackathon for Social Good in the Arab World. Abu Dhabi 25/4/2019
26. **Lexicographic Search Engine** ([news](#)). *Winner Lightning Talk*. Ceremony of the Mohammed Bin Rashid Award for Arabic Language. Dubai 11/4/2019
27. **Building a Big Linguistic Data Graph for Arabic Natural Language Processing** ([program](#)). *Featured Keynote*. The 6th Arab Women in Computing Annual International conference. Rabat, Morocco 8/3/2019
28. **The Arabic Ontology and Lexicons** ([video](#) - [media](#)). *Featured Keynote*. Renaissance Forum - The Arab Renaissance for Democracy and Development. Amman, Jordan 21/2/2019
29. **Eight Years of Research Make Its Debut** ([ceremony](#) - [media](#) - [media](#)). *Featured Keynote*. The Launching Ceremony of the Arabic Ontology and the Lexicographic Search Engine. Birzeit University. Palestine 25/9/2018
30. **Search Engine for Arabic Lexicons** ([article](#) - [program](#)). *Research Talk*. The 5th Conference on Translation and the Problematics of Cross-cultural Understanding. The Forum for Arab and International Relations. Qatar. 12/12/2018
31. **The Arabic Ontology and Lexical Semantic Resources** ([program](#)). *Keynote Speaker*. The INFOL@NGUES 2018 Conference Beja, Tunisia 2018
32. **Classifying Processes and Basic Formal Ontology** ([slideshare](#) - [slides](#)). *Research Talk*. The International Conference on Biomedical Ontology (ICBO). Newcastle, UK 2017
33. **Clustering Arabic Tweets for Sentiment Analysis** ([slideshare](#)). *Research Talk*. 2017 IEEE/ACS 14th International Conference on Computer Systems and Applications. IEEE Computer Society. Hammamet, Tunisia 2017
34. **Arabic Ontology Engineering** ([tutorial](#)). *Tutorial*. The 2017 IEEE/ACS 14th International Conference on Computer Systems and Applications. IEEE Computer Society. Hammamet, Tunisia 2017
35. **e-Government for Better Migration Governance - The Status in Palestine, Jordan and Lebanon**. *Invited Talk*. The Sub-Regional Workshop on Migration Governance, EUROMED Migration IV. Amman, Jordan 11/10/2017
36. **Medical Terminology in the Arabic Ontology** *Invited Talk*. Hi-Cure workshop. Lisbon Portugal 12/4/2016
37. **Zinnar - The e-Government Interoperability Framework** ([slides](#) - [Zinnar](#) - [Summit Page](#)). *Keynote Speaker*. Summit On ICT-enabled Governance, and the The 4th International Summer School On Open and Collaborative Governance. Samos, Greece 2015
38. **Government Semantics For Data and Services** ([video P1](#) - [P2](#)). *Invited Talk*. Government Course, Tallinn University of Technology. Estonia 29/9/2015
39. **Advances in ICT**. *Panel Chair*. The First National Forum for Innovators - Higher Council for Innovation & Excellence, Palestine 12/9/2015
40. **The Arabic Ontology**. *Invited Talk*. American University of Beirut, Lebanon 3/9/2015

41. **On Computer Science Trends and Priorities in Palestine** ([slideshare](#) - [Slides](#)). *Keynote Speaker*. Workshop on IT Research Trends and Priorities. The Islamic University of Gaza. Palestine 28/3/2015
42. **Lessons Learned from Class Recording and Publishing of Eight Online Courses** ([Slides](#)). *Invited Talk*. Workshop on e-learning in Higher Education. Birzeit University. Palestine 6/12/2014
43. **Future Internet in Horizon 2020 Calls** ([slides](#)). *Keynote Speaker*. The 1st ICT Networking Event. Amman, Jordan 13/10/2014
44. **Fund-raising and Participation Opportunities in EU Programs**. *Invited Talk*. Workshop on Fund-raising and Participation in EU projects. Palestinian Technical University, Tulkarim 3/9/2014
45. **Towards Developing an e-Governance Interoperability Framework at Palestinian Universities**. *Keynote Speaker*. Workshop on e-Governance at Palestinian Universities. Al-Quds Open University. Tulkarim 1/6/2014
46. **Recommendations for Writing a Comparative Proposal and Networking with International Scientists**. *Invited Talk*. Workshop on Fund raising and Participation in EU projects. Arab American University in Jenin. Palestine 12/2/2014
47. **Web 3.0 and Semantic Web**. *Keynote Speaker*. The Tech-con2 - Developers Day. Ramallah, Palestine 12/10/2013
48. **Ontologizing Dictionaries and Lexical Database** ([program](#)). *Keynote Speaker*. The Arab-Term Workshop, at the German-Jordanian University. Jordan 15/9/2013
49. **Building a Formal Arabic Ontology** ([article](#) - [program](#)). *Research Talk*. The 2nd International conference on Arabic Language. UAE 8/5/2013
50. **Writing a Competitive Proposal Highlights from the e-Government Academy Project** ([video1](#) - [video2](#)). *Keynote Speaker*. The 2013 EU Tempus Information Day. Ramallah, Palestine 16/1/2013
51. **Europe-Palestine Research Cooperation —Ongoing Projects at Sina Institute at Birzeit University** ([video](#)). *Keynote Speaker*. The 2nd International Platform on Integrating Arab e-Infrastructure in Global Environment (e-AGE), Arab States Research and Education Network (ASREN). Dubai 13/12/2012
52. **Zinnar - The e-Government Interoperability Framework** ([video](#) - [program](#)). *Keynote Speaker*. The First National Conference on e-Governance and e-Services. Palestine 27/6/2012
53. **Building a Formal Arabic Ontology** ([article](#) - [slides](#)). *Invited Talk*. The Experts Meeting on Arabic Ontologies and Semantic Networks. ALECSO, Arab League. Tunisia 26-28/6/2011
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55. **Ontology-Based Data and Process Governance Framework —The Case of e-Government Interoperability in Palestine** ([article](#) - [slides](#)). *Research Talk*. The IFIP International Symposium on Data-Driven Process Discovery and Analysis (SIMPDA). Campione, Italy 30/6/2011
56. **Ontology-Based Data Governance** ([article](#) - [slides](#)). *Invited Talk*. The eGovernance Workshop, at the Ministry of Telecommunication and Information Technology. Palestine 28/6/2011
57. **Arabic Ontology Engineering —Research Challenges and Opportunities** ([article](#) - [conference](#)). *Keynote Speaker*. The International Conference on Intelligent Semantic Web -Services and Applications (ISWSA). Jordan 18/4/2011
58. **eGovernment Interoperability Framework in Palestine**. *Invited Talk*. at the Service Oriented Architecture Course, Masters on eGovernment. University of Trento. Italy December, 2010
59. **Building an Arabic Ontology, for Information Search and Integration**. *Invited Talk*. ExpoTech III, Birzeit University. Palestine 16/4/2010
60. **Web 3.0, The Third Generation of the Web**. *Invited Talk*. ExpoTech III, Birzeit University. Palestine 16/4/2010

61. **Mechanisms of Connecting IT Colleges in the Fields of Research and Teaching.** *Invited Talk*. Workshop on Evaluation of IT curriculum in Palestinian universities. Arab American University of Jenin. Palestine 28/3/2010
62. **Legal Ontology Engineering.** *Tutorial*. Institute of Law, Birzeit University. Palestine 2009
63. **Ontology-Based e-Governance.** *Invited Talk*. e-government Course. Birzeit University. Palestine 17/12/2009
64. **A Data Mashup Language for the Data Web.** *Invited Talk*. IFIP 2.6 Research Workshop. Nicosia, Cyprus 24/4/2009
65. **A Data Mashup Language for the Data Web.** *Research Talk*. LDOW workshop at the WWW Conference. Madrid, Spain 20/4/2009
66. **MashQL: A Query Formulation and Mashup Language.** *Research Talk*. University of Cyprus. Cyprus 25/2/2009
67. **Towards Data Mashups and Pipes.** *Research Talk*. ONISW workshop at the CiKM Conference. Napa Valley, California, USA 30/10/2008
68. **MashQL: Towards Data Mashups and Pipes.** *Invited Talk*. University of Athens. Greece 21/10/2008
69. **Web 3.0 Information Systems.** *Invited Talk*. The "Research and Technology Course (CS671). University of Cyprus. Cyprus 19/2/2008
70. **Information Search.** SEARCHiN Kick-off Meeting. *Research Talk*. University of Cyprus. Nicosia, Cyprus 10/12/2007
71. **Semantic Pipes -Towards Effective Information Search.** SEARCHiN Kick-off Meeting). *Research Talk*. University of Cyprus. Cyprus 10/12/2007
72. **Web 2.0, Social Programming, Mashups -What is in it for me!.** *Research Talk*. University of Cyprus. Cyprus 6/12/2007
73. **Formalizing ORM using the DLR and the SHOIN/OWL Description Logics.** *Research Talk*. The International Workshop on Object-Role Modeling (ORM). Algarve, Portugal 28/11/2007
74. **Reasoning on ORM Schemes -Formalizing ORM using DLR and SHOIN/OWL.** *Research Talk*. The International Conference on Conceptual Modeling (ER). Auckland, New Zealand 8/11/2007
75. **Ontology Engineering —Challenges and Opportunities.** *Invited Talk*. University of Cyprus. Cyprus 13/9/2007
76. **ORM and Description Logic.** *Invited Talk*. Web Information Systems Course. University of Hasselt, Belgium 19/4/2007
77. **Reasoning on ORM Schemes.** *Research Talk*. The Dutch-Belgian Database Day. Brussels, Belgium 15/11/2006
78. **Towards the Adoption of Linguistic Resources in Formal Ontology Engineering.** *Invited Talk*. IFIP 2.12/12.4 Research Meeting. Montpellier, France 2/11/2006
79. **The Ontology Outreach Advisory.** *Research Talk*. The OOA Kick-off workshop on Semantic challenges and opportunities in the Human Resources Domain - As part of the ePortfolio Conference. Oxford, England 11/10/2006
80. **Towards the Notion of Gloss, and the Adoption of Linguistic Resources in Formal Ontology Engineering.** *Research Talk*. The WWW Conference, Semantic Web Track. Edinburgh, UK 26/5/2006
81. **Unsatisfiability Reasoning Patterns in ORM Conceptual Schemes.** *Research Talk*. The International Conference on Semantics in Networked Word (ICSNW), Munich, Germany 29/3/2006
82. **Modularization and Auto-Composition of ORM Schemes.** *Research Talk*. The International Workshop on Object-Role Modeling (ORM). Larnaca, Cyprus 3/11/2005
83. **The Double Articulation Principle for Ontology Engineering.** *Research Talk*. University of Trento, Italy 31/8/2005
84. **Towards Methodological Principles for Ontology Engineering.** *PhD Public Defense*. PhD Public Defense. Vrije Universiteit Brussel, Belgium 26/5/2005

85. **Ontology Modularization and Composition.** *Research Talk.* Workshop on Ontology Modularization and Context. Vrije Universiteit Brussel, Belgium 14/10/2004
86. **The Dogma Methodology for Ontology Engineering.** *Invited Talk.* IFIP 2.6 Research Workshop. Sicily, Italy. 7/11/2003
87. **Ontology-Based Customer Complaint Management.** *Research Talk.* Workshop on Regulatory Ontologies and the Modeling of Complaint Regulations (WORM). Catania, Sicily, Italy 4/11/2003
88. **Formal Ontology Engineering —The DOGMA Approach.** *Research Talk.* The International Conference on Ontologies, Databases and Applications of Semantics (ODBase). Irvine, California, USA 31/10/2002
89. **A Conceptual Markup Language that Supports Interoperability between Business Rule Modeling Systems.** *Research Talk.* The 10th International Conference on Cooperative Information Systems (CoopIS). Irvine, California, USA 31/10/2002
90. **A Markup Language for ORM Business Rules.** *Research Talk.* The International Workshop on Rule Markup Languages for Business Rules on the Semantic Web, (RuleML). Sardinia, Italy 14/6/2002
91. **Decomposing Ontology into an Ontology Base and its Interpretation Layer.** *Research Talk.* Dagstuhl Seminar N° 02061: Rule Markup Techniques for the Semantic Web. Dagstuhl, Germany 3/2-8/2/2002

In the Media

reports and interviews about my research

1. **Interviews about the the Arabic Dialects Corpus** ([Al-Jazeera](#) • [AjyaL](#) • [Independent](#) • [more..](#)) 2023
2. **On Wiki Palestine and the Arabic Content on the Web** ([24FM](#)) December, 2022
3. **On the World Arabic Language Day** ([Palestine TV](#)) December, 2019
4. **Prof. Jarrar wins the Shoman Arab Researchers Award** ([Palestine TV](#) • [Quds TV](#) • [Ajjyal](#) • [Ramallah](#) • [25 more..](#)) September, 2019
5. **Arabic and AI: Why voice-activated tech struggles in the Middle East** ([Middle East Eye](#)) September, 2019
6. **Birzeit University won the Bin Rashid Arabic Language Award for its Lexicographic Search Engine** ([30 newspapers..](#)) April, 2019
7. **On DNA data storage** ([France 24](#)) April, 2019
8. **On the International Mother Language Day** ([Roya TV](#)) February, 2019
9. **Birzeit University symposium details efforts to create Arabic search engine** ([15 newspapers..](#)) February, 2019
10. **An Arabic Greeting to Birzeit University** ([Echorouk online](#) • [others](#) [8 newspapers..](#)) December, 2018
11. **An Online Arabic Dictionary Makes Its Debut** ([Al-Fanar Media](#) • [20 newspapers..](#)) November, 2018
12. **Reportage about the Arabic Ontology** ([Al Arabiya TV](#)) September, 2018
13. **Birzeit announces linguistic search engine and the Arabic Ontology** ([Palestine TV](#) • [Palestine TV report](#) • [Quds TV](#) • [FM24](#) • [35 more..](#)) September, 2018
14. **Birzeit signs MoU with ALESCO to digitize and integrate Arabic lexicons** ([Ajjyal](#) • [ALECSO](#) • [10 more..](#)) December, 2017

15. **Launching the Corpus for Palestinian Dialect (Curras)** ([Wamda](#) • [Palestine Today](#) • [Al-Iqtisadi](#) • [10 more..](#)) January, 2017
16. **UB Department of Biomedical Informatics hosts its first Fulbright scholar** ([UB](#) • [BZU](#)) October, 2016
17. **MoU with RedCrow to Cooperate in Computational Linguistics** ([press](#) • [5 more..](#)) May, 2016
18. **Interviews about digitization and engineering of dictionaries** ([Palestine Today TV](#) • [Ajyyal](#) • [9 more..](#)) March, 2016
19. **Prof. Jarrar wins Google Research Award** ([Palestine Voice](#) • [NisaFM](#) • [24FM](#) • [35 more..](#)) February, 2016
20. **Deploying a SMART Multilingual Audio Guide in Bethlehem** ([Nisa FM](#) • [Sina](#) • [40 more..](#)) August, 2014
21. **Birzeit University won an EU-FP7 project in ICT Industry** ([15 newspapers..](#)) October, 2013
22. **Prof. Fellbaum from Princeton Concluded her Visit to Sina Institute** ([press](#) • [10 more..](#)) July, 2013
23. **Celebrating the End of the e-Government Training Program** ([ceremony](#) • [8 newspapers](#)) September, 2012
24. **Interviews about the First National Conference on e-Governance and e-Services** ([Maan TV 1](#) • [Palestine TV 1](#) • [Maan TV 2](#) • [Palestine TV 2](#) • [RayaFM](#) • [Ajyyal](#) • [40 more..](#)) June, 2012
25. **Birzeit Wins an EU-FP7 Project on Multilingual Knowledge Sharing Technologies** ([press](#) • [ceremony](#) • [Palestine TV](#) • [Kickoff](#) • [8 newspapers](#)) December, 2011
26. **Launching the Palestinian e-Government Interoperability Framework** ([ceremony](#) • [more..](#)) 2011